

Dialectica for Constructible Falsity

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Abstract

We can instantiate the Dialectica construction to the category of Heyting algebras, as the category of Heyting algebras has all finite products and pullbacks. Then we get not only a model of classical linear logic, as it happens for other categories with finite limits, but we also get a model of a different logic, related to Nelson’s ‘Constructible Falsity’. The Chu/Dialectica construction considered over Heyting algebras provides us with a bridge to ‘strong negation’ based logics. We discuss how, while the original idea was basically described in Patterson’s 1998 doctoral thesis, much work on Nelson’s systems was and still is required to make good on the claim that the categories can be considered models for these different logics. We also discuss briefly why this is important.

Dedication

This short note is dedicated to Andrea Loparic, a Brazilian logician recently departed and one of the heroines of Logic in Brazil. The original plan was doing some automated logic work with Andrea, but these were curtailed by her passing. Andrea (with her son Marko Loparic) had a system for proving intuitionistic theorems online. For a long time, I wanted to use their system, and especially I wanted to enroll her expertise, acquired by teaching logic from Kleene’s book [Kle52] for decades, to use automated theorem proving in a different way. More than simply proving the well-known intuitionistic theorems in Kleene’s book, I wanted to see how many different ways could we prove them true, given some restrictive assumptions. For example, what if we assumed a linear or a relevant use of hypotheses? Which one of Kleene’s basic intuitionistic theorems remains true? If they remain true, how many different proofs in sequent calculus can we find of a given theorem and how do we characterize the differences between proofs? I have not had the chance to work with Andrea on this problem. Together with Elaine Pimentel, Giselle Reis, and Carlos Olarte, we started investigating some of these questions [OdPPR18]. But time is of essence in a commemorative volume, so we have decided to write down another strand of research, an application of Heyting algebras in the Dialectica construction, for this volume honoring Andrea.

This corresponds to a simplification of the setting of the Dialectica categories to the universe of Heyting algebras, the traditional models for propositional intuitionistic logic. I hope Andrea would approve of this application. The setting of Heyting algebras makes the Dialectica construction less abstract, so we can see more clearly where certain assumptions are needed. The surprise is that this setting can also be seen as a model for a variant of Nelson’s Constructible Falsity system, which always seemed to me an interesting, but flawed logic system. While the intuition behind Constructible Falsity seems very elegant, the logic does not satisfy the substitution of equals by equals, as discussed, for example by Odintsov in [Odi04]. This seems a big problem, even if there are many interesting algebraic properties proved for the system.

1 Introduction

In this note, we describe a simplified version of the Dialectica construction [dP89b], done not over the category **Set** of sets nor over other more general category **C**, but over the specific category **Hey** of Heyting algebras.

The Dialectica construction [dP91b] is a categorical logic construction, originally based on Gödel’s Dialectica Interpretation [GF⁺86]. The construction comes in two different ‘flavors’:

- (i) an intuitionistic linear logic one, that nowadays it is called **Dial** (as this is closer to the functionals described in Gödel's Dialectica interpretation), published in [dP89a] and
- (ii) a classical (or full intuitionistic) linear logic one, that used to be called **GSet**, G for Jean-Yves Girard, who suggested it as a way to provide models of classical linear logic [dP89b]. This is now called **LDial**, for Linear Dialectica.

Apart from the conference versions cited, both variants of the Dialectica construction are only described in the technical report of the doctoral thesis [dP91b].

The construction can be made using any category $\mathbb{C}$ with finite limits (products and pullbacks are used), but more structure is required to model the connectives of the logic. We usually describe the construction using sets, as the notion of a map in this category can be hard to grasp. The next section describes the linear version of the Dialectica construction we will be using. Then we describe the category of Heyting algebras **Hey** and show that it is a cartesian closed category, so we can make use of the Dialectica construction over it.

2 Dialectica constructions

The main idea of the **original** Dialectica construction is to build a category of **relations** between objects of a basic category $\mathbb{C}$, where $\mathbb{C}$ has all finite limits. So if U and X are objects of $\mathbb{C}$, a typical object of $\text{Dial}_2(\mathbb{C})$ will be $A \xrightarrow{\alpha} U \times X$, a sub-object of the product object $U \times X$. Or an equivalence class of monics into the product $U \times X$. Recall that ‘relations’ in category theory can be represented either as subobjects $A \xrightarrow{\alpha} U \times X$ or as maps into some sort of classifier like **2**. Correspondingly, we write an object A as either (U, X, α) or $A \xrightarrow{\alpha} U \times X$ or still as $U \times X \rightarrow \mathbf{2}$, assuming that the category $\mathbb{C}$ has an element $2 = 1 + 1$, corresponding to truth-values $\{\text{true} = 1, \text{false} = 0\}$.

These objects intuitively represent formulae of the form $\exists U \forall X. \alpha(u, x)$. Thus they are not symmetric and we can think of them as games between a prover $\exists$ loise and a refuter $\forall$ belard. But more often than not we think of them as instances of problems and possible solutions to these problems, where the relation α means that the instance of the problem is solved (or not) by a given instance of the solution.

A map between two such objects $A \xrightarrow{\alpha} U \times X$ and $B \xrightarrow{\beta} V \times Y$ is a pair of maps from $\mathbb{C}$, (f, F) , $f : U \rightarrow V$ and $F : U \times Y \rightarrow X$, such that a non-trivial condition is satisfied. This condition is logically described in the definitions below. But the nature of the morphisms is that they correspond to some sort of implication between these games or pairs problem/solution.

This construction can be summarized in the definition below:

Definition 2.1 (Original Dialectica category) *Given a category $\mathbb{C}$ with finite limits, the category $\text{LDial}_2(\mathbb{C})$ has as objects triples of the form (U, X, α) sometimes written as $(U \xleftarrow{\alpha} X)$, where U and X are objects of $\mathbb{C}$ and $\alpha \subseteq U \times X$ is a ‘relation’, that is a monomorphism into $U \times X$. A map between two objects $(U \xleftarrow{\alpha} X)$ and $(V \xleftarrow{\beta} Y)$ is a pair of maps from $\mathbb{C}$, (f, F) , $f : U \rightarrow V$ and $F : U \times Y \rightarrow X$, such that a non-trivial logical condition is satisfied, namely that*

$$\text{For all } u \in U \text{ and } y \in Y, u\alpha F(u, y) \text{ implies } f(u)\beta y$$

The categorical version of this definition says that pulling back the sub-object $A \xrightarrow{\alpha} U \times X$ along the map $\langle \pi_1, F(u, y) \rangle : U \times Y \rightarrow U \times X$ and the sub-object $B \xrightarrow{\beta} V \times Y$ along the map $f \times \text{id}_Y : U \times Y \rightarrow V \times Y$ as the diagram below shows, the first sub-object A' is smaller than the second sub-object B' , i.e. there is a map $A' \rightarrow B'$ making the triangle below commute:

$$\begin{array}{ccccc}
& & A' & \longrightarrow & A \\
& \swarrow & \downarrow & & \downarrow \alpha \\
B' & \longrightarrow & U \times Y & \xrightarrow{\langle \pi_1, F \rangle} & U \times X \\
\downarrow & & \downarrow f \times Y & & \\
B & \xrightarrow{\beta} & V \times Y & &
\end{array}$$

But in this short note, we will be interested instead on the **Linear** Dialectica category, called when it first appeared, a Dialectica-like category [dP89b]. This was suggested by Girard to simplify the original Dialectica construction, as a way of producing categorical models of Linear Logic, not only Intuitionistic Linear Logic (ILL). Also in what follows, we will prefer to think of the objects of this category as maps into an algebraic structure called a *lineale*. Lineales are poset reflections of symmetric monoidal closed categories, hence they are algebras with multiplicative operations tensor, negation, and linear implication, see [dP02]. Clearly, the lineale $\mathbf{2}$, consisting of the objects $\{0, 1\}$ meant as $0 = \text{false}$ and $1 = \text{true}$, is a specific instance of the construction below.

Definition 2.2 (Linear Dialectica category) *Given a category $\mathbb{C}$ with finite limits, the linear dialectica category $\mathbf{LDial}_L(\mathbb{C})$ has as objects triples of the form $(U \multimap^\alpha X)$, where U and X are objects of $\mathbb{C}$ and α a ‘multirelation’ between U and X is a map $\alpha : U \times X \rightarrow L$, where L is a lineale. A map between two objects $(U \multimap^\alpha X)$ and $(V \multimap^\beta Y)$ is a pair of maps from $\mathbb{C}$, (f, F) , $f : U \rightarrow V$ and $F : Y \rightarrow X$, such that in the following diagram*

$$\begin{array}{ccc}
U \times Y & \xrightarrow{U \times F} & U \times X \\
f \times Y \downarrow & \geq & \downarrow \alpha \\
V \times Y & \xrightarrow{\beta} & \mathbf{2}
\end{array}$$

for all $u \in U$ and $y \in Y$, $u \alpha F(u, y) \leq f(u) \beta y$ in L .

We could go on and describe other categorical structures available in the category $\mathbf{LDial}_L(\mathbb{C})$, used to model the additives and especially the modalities of Linear Logic [dP91a], but we prefer to describe the category $\mathbb{C}$ we interested in, the category of Heyting algebras.

3 Heyting Algebras

Heyting algebras are the algebras used to model propositional intuitionistic logic. So they correspond to Boolean algebras for constructive propositional logic. One could define a Heyting algebra $\mathbf{H}$ as a bounded lattice with all exponential objects. Bounded means our lattice has a top $\top$ and a bottom element $\perp$. The lattice $\mathbf{H}$ is regarded as a category where meet (denoted as $\wedge$) is the categorical product. The exponential condition means that for any objects B and C of $\mathbf{H}$ an exponential C^B exists and is unique.

A Heyting implication (often written using $B \rightarrow C$) is just an exponential, an alternative notation for C^B , used when we are talking about a cartesian category. From the definition of exponentials, we have that implication $- \rightarrow - : H \times H \rightarrow H$ is right adjoint to meet $\wedge : H \times H \rightarrow H$. This adjunction can be written as

$$(- \wedge B) \dashv (B \rightarrow -)$$

or, if we write **Hey** for the category of Heyting algebras and their morphisms, we have that

$$\text{Hom}_{\text{Hey}}(A \times B, C) \cong \text{Hom}_{\text{Hey}}(A, B \rightarrow C)$$

Many equivalent definitions of a Heyting algebra can be given.

Definition 3.1 *A Heyting algebra is a residuated lattice whose monoid operation is the meet $\wedge$ of the lattice.*

The monoid unit must then be the top element $\top = 1$. The commutativity of this monoid implies that the two residuals coincide as $A \rightarrow B$.

Some examples of Heyting algebras:

1. Every Boolean algebra is a Heyting algebra, with $A \rightarrow B$ given by $\neg A \vee B$.
2. Every totally ordered set that has a least element 0 and a greatest element 1 is a Heyting algebra (if viewed as a lattice). In this case $A \rightarrow B$ equals to 1 when $A \leq B$, and B otherwise.
3. The simplest Heyting algebra that is not already a Boolean algebra is the totally ordered set $\{0, 1/2, 1\}$ (viewed as a lattice) with operations to make it so see https://en.wikipedia.org/wiki/Heyting_algebra.

Morphisms of Heyting algebras should preserve both the distributive lattice structure and the logical implications (internal-homs). More formally:

Definition 3.2 *Given two Heyting algebras H_1 and H_2 and a mapping $f : H_1 \rightarrow H_2$, we say that f is a morphism of Heyting algebras if, for any elements A and B in H_1 , we have:*

$$\begin{aligned} f(\perp) &= \perp \\ f(A \wedge B) &= f(A) \wedge f(B) \\ f(A \vee B) &= f(A) \vee f(B) \\ f(A \rightarrow B) &= f(A) \rightarrow f(B) \end{aligned}$$

Using any of the equivalent definitions of Heyting algebra and the notion of morphism above we can define the category **Hey of Heyting algebras**. Heyting algebras are always **distributive**. Specifically, we always have the identities:

$$\begin{aligned} A \wedge (B \vee C) &= (A \wedge B) \vee (A \wedge C) \\ A \vee (B \wedge C) &= (A \vee B) \wedge (A \vee C) \end{aligned}$$

The following infinite distributive law holds in any **complete** Heyting algebra:

$$A \wedge \bigvee Y = \bigvee \{A \wedge y \mid y \in Y\}$$

for any element A in H and any subset Y of H . Conversely, any complete lattice satisfying the above infinite distributive law is a complete Heyting algebra, with

$$A \rightarrow B = \bigvee \{C \mid A \wedge C \leq B\}$$

being its relative pseudo-complement operation.

Given the adjunction that holds for Heyting algebras, it is clear that they form a (poset-like) (bi)cartesian closed category, hence we can use the Dialectica category construction over this category. Heyting algebras have finite products and coproducts, and products distribute over coproducts.

4 Dialectica over Heyting

Instantiating the dialectica construction above to the category of Heyting algebras $\mathbf{Hey}$ and the lineale 2 , we have:

Definition 4.1 *Objects of $\mathbf{LDial}_2(\mathbf{Hey})$ are relations $\alpha : H \times J \rightarrow 2$ between two Heyting algebras H and J , hence triples (H, J, α) . Morphisms between two objects (H_1, J_1, α) and (H_2, J_2, β) are pairs of Heyting algebra morphisms $(f, f\#)$ where $f : H_1 \rightarrow H_2$ and $f\# : J_2 \rightarrow J_1$ make the dialectica condition hold. This means for all h in H_1 and all j in J_2 , we have that $\alpha(h, f\#(j)) \Rightarrow \beta(f(h), j)$.*

Composition in the second coordinate is not trivial, but a simple calculation allows us to state:

Proposition 4.2 *Given the category $\mathbf{Hey}$ of Heyting algebras, we can perform the dialectica construction over it, obtaining the category $\mathbf{LDial}_2(\mathbf{Hey})$.*

As far as the categorical structure of $\mathbf{LDial}_2(\mathbf{Hey})$ is concerned, we should have a monoidal closed structure, given by the internal-hom that internalizes the morphisms of $\mathbf{LDial}_2(\mathbf{Hey})$ and the cross-product tensor, $\bowtie : \mathbf{Dial}(\mathbf{Hey}) \times \mathbf{Dial}(\mathbf{Hey}) \rightarrow \mathbf{Dial}(\mathbf{Hey})$, which is left adjoint to it.

Definition 4.3 *Given objects (H_1, J_1, α) and (H_2, J_2, β) of $\mathbf{Dial}(\mathbf{Hey})$, their internal-hom is given by the object $(H_2^{H_1} \times J_1^{J_2}, H_1 \times J_2, [\alpha, \beta])$, where the relation $[\alpha, \beta]$ is given by evaluating the function spaces and then multiplying the relations $\alpha : H_1 \times J_1 \rightarrow 2$ and $\beta : H_2 \times J_2 \rightarrow 2$.*

Note that this leaves us with a map $H_2^{H_1} \times J_1^{J_2} \times H_1 \times J_2 \rightarrow H_2 \times J_1 \times H_1 \times J_2 \rightarrow 2 \times 2$, but we can use the implication in $2 \times 2 \rightarrow 2$ to get the right codomain.

This internal-hom is simply the internalization of the simplified morphisms of $\mathbf{Dial}_2(\mathbf{Hey})$. However, to obtain the right tensor product, we need to reverse engineer from the desired adjunction.

Definition 4.4 *Given objects (H_1, J_1, α) and (H_2, J_2, β) of $\mathbf{Dial}(\mathbf{Hey})$, their cross-product tensor product is given by $(H_1 \times H_2, J_1^{H_2} \times J_2^{H_1}, \alpha \bowtie \beta)$. The relation $\alpha \bowtie \beta$ is given by*

Since we reverse-engineered our tensor product it is easy to see that we have a symmetric monoidal closed structure in $\mathbf{Dial}(\mathbf{Hey})$.

Theorem 4.5 *The category $\mathbf{LDial}_2(\mathbf{Hey})$ is symmetric monoidal closed when considered with the cross-product tensor $\bowtie$ and the natural internal-hom $[-, -]_{\mathbf{LDial}_2(\mathbf{Hey})}$.*

(There is another tensor product in the category $\mathbf{LDial}_2(\mathbf{Hey})$, which is much simpler, we simply multiply the relations, but this tensor has no internal-hom as far as I can discern.)

Now because any category resulting from the dialectica construction is a model of linear logic (without modalities, for the time being), this means that $\mathbf{LDial}_2(\mathbf{Hey})$ is a model of linear logic. We have bifunctors that model the multiplicatives $\otimes = \bowtie, [-, -] = (- \multimap -), \wp$. Negation is simply transposing the Heyting algebras and negating the relation in (H_1, J_1, α) to get $(J_1, H_1, \neg\alpha)$.

Theorem 4.6 *The category $\mathbf{LDial}(\mathbf{Hey})$ is a model of classical linear logic, without the modalities $!, ?$.*

The proof can be found in its more general form in [dP89b]. But further work, modeling the modalities, can transform this model of linear logic into a model of intuitionistic logic, an important feature of linear logic.

The generic transformation previously described requires a considerable amount of structure: free monoids in the base category, as well as two comonads and their composition, together with co-algebras and co-Kleisli categories for the comonads. We refrain from attempting to prove that the category of Heyting algebras satisfies the necessary conditions, as an easier pair of modalities can be described.

5 Predicate-free Dialectica

So far we provided a simple instantiation of what we had many years ago. The only difference is that now we are taking the category $\mathbf{C}$ as the category of Heyting algebras $\mathbf{Hey}$. However, we now can think about the dialectica objects themselves, that is the relations $U \times X \rightarrow L$, and we can simplify these too.

If instead of the object of truth-values 2 as we have been taking our relations into, we can decide to have maps into the lineale 1 , i.e. some sort of degenerated relations. Since 1 is the terminal object, there is only one such unary relation, even for a generic category $\mathbf{C}$. Moreover, the dialectica condition is always satisfied, as mapping into the terminal object 1 the previous square that we required to commute, will commute all the time, as all maps from $U \times Y$ into 1 are the same map. This construction is very degenerated. Morphisms still have to be maps of the right domain and codomain, but the dialectica condition is satisfied automatically. Hence we are talking about Chu spaces [Pra99]. The Chu condition that requires maps $(f, f\#) : U \rightarrow V, Y \rightarrow X$ to be equal over the proposed lineale L is vacuously satisfied when $L = 1$.

Definition 5.1 *Given a dialectica category $\mathbf{LDial}_L(\mathbf{C})$, we call the instantiation of the lineale L to the lineale 1 , $\mathbf{LDial}_1(\mathbf{C})$ the predicate-free dialectica construction.*

Since $\mathbf{C}$ has finite limits, it has a terminal object, so the predicate-free dialectica construction is always available and may appear somewhat boring. After all, we intend to do some logic, but then throw it all away, by making true the same as false, and collapsing all the connectives in the object 1 . But this is exactly what one does when discussing **compact closed categories** and some of these are very useful.

We remark that the instantiation of the lineale L to 1 can be done also in the original dialectica construction. In that case, instead of the Chu construction, one obtains **lenses**, or **containers**, but these seem to be more useful if one generalizes the system to using dependent types. There are many applications of (dependent) lenses to databases and other areas of computing.

The linear dialectica category $\mathbf{LDial}_1(\mathbf{C})$ is both a dialectica category and a Chu space. The original dialectica category $\mathbf{Dial}_1(\mathbf{C})$ consists of monomorphic lenses, which can be thought of as a degenerate (non-dependent version of a) polynomial category.

There is a much more intricate version of a dialectica-like category [MvG18], for a more expressive version of logic, dependent type theory, where morphisms are similar to the ones we have described in the original dialectica construction. Sean Moss and Tamara von Glehn in [MvG18], describe a fibration version of a dialectica-like category, where the base category is a model of **dependent type theory** and the total category is another model of dependent type theory. In that construction, predicates are simplified to $\top$, so it is also a category of polynomials. Polynomials are ubiquitous in mathematics and several works describe categorical versions of polynomials, including the recent book [SN]. Further categorical similarities and differences between polynomial functors and original dialectica categories are discussed by Niu in [Niu] and by Capucci et al [CGM⁺24]. This comparison is based on the work of Sean Moss and Tamara von Glehn [Mos22] and Pieter Hofstra’s [Hof11]. This shows generalizations of the (fibrational) Dialectica construction and of polynomials, the so-called ‘Dialectica polynomials’. These polynomials are given by the coproduct completion of the product completion $\Sigma\Pi(\mathbf{C})$ of a category $\mathbf{C}$.

Similarities and differences between the categorical structures of Chu categories and those of the linear dialectica categories were initially discussed in [dP06]. There it is shown how a seemingly small change in the definition of a morphism can have a large effect on the categorical structure of the whole category. In particular, we can see that the easy tensor product of the original dialectica category, which also exists in the linear dialectica construction, is not a left-adjoint to the internal-hom anymore. We need to consider a much more complicated, ‘crossing’ tensor product to show a monoidal closed structure when we move to the setting of the linear dialectica category.

More recently, Shulman [Shu18] described a generalization of both dialectica and Chu spaces, the 2-Chu-Dialectica construction over polycategories, which allows us to discuss 2-categorical dualities and has some nice connections to multivariable adjunctions. These higher-order categorical constructions are fascinating. However, in this note we discuss a different kind of connection, a logical one to David Nelson’s logics of strong negation and constructible falsity. We first recall Nelson’s systems.

6 Nelson’s Systems

David Nelson introduced a number of systems of non-classical logics in the papers [Nel49, Nel59, AN84] providing for more fine-grained analyses of notions such as ‘falsity’ and ‘negation’ than either classical or intuitionistic logic can afford. His resulting logics combine an intuitionistic approach to truth with a dual-intuitionistic treatment of falsity. His systems were investigated algebraically and using proof-theoretical tools (mainly sequent-calculus and natural deduction) in many publications. A comprehensive survey is found in [KW12].

Nelson’s systems strengthen the concept of negation in intuitionistic logic in that strong negation is somewhat constructive: it produces counterexamples for negated propositions. Nelson’s analysis of the meaning of ‘falsity’ is analogous to the intuitionistic analysis of ‘truth’. The property advocated by Nelson – namely, if a formula $\neg(A \wedge B)$ is provable, then either $\neg A$ or $\neg B$ is provable – is one that may be regarded as a dual to the well-known *disjunction property* of intuitionistic logic. This property is what Nelson calls ‘constructible falsity’.

Considering only the propositional fragment of Nelson’s systems, the language of the logics $\mathcal{N}3$ and $\mathcal{N}4$ comprises a conjunction ($\wedge$), a disjunction ($\vee$), a ‘strong’ negation ($\sim$) and two implications — a ‘weak implication’ ($\rightarrow$) and a ‘strong implication’ ($\Rightarrow$), usually introduced via the following term: $A \Rightarrow B := (A \rightarrow B) \wedge (\sim B \rightarrow \sim A)$. The presence of two implications is crucial in Nelson’s logics. The strong implication in $\mathcal{N}3$ has the feature that $(p \wedge \sim p) \Rightarrow q$ is not provable, while the weak implication satisfies the more usual $(p \wedge \sim p) \rightarrow q$. Meanwhile, the weak implication ($\rightarrow$) allows us to see $\mathcal{N}3$ and $\mathcal{N}4$ as conservative extensions of positive intuitionistic logic, but the strong implication ($\Rightarrow$) permits us to view their algebraic counterparts as residuated structures, and therefore to regard $\mathcal{N}3$ and $\mathcal{N}4$ as strengthenings of well-known substructural or relevance logics, as explained in [NRMS20].

Nelson’s logics share many properties with the positive fragment of intuitionistic logic (in particular, they do not validate Pierce’s law $((p \Rightarrow q) \Rightarrow p) \Rightarrow p$). According to Kamide and Wansing [KW12], page 2, a Hilbert-style system for $\mathcal{N}3$ is obtained from that of $\mathcal{N}4$ by adding the axiom scheme $(p \wedge \sim p) \rightarrow q$. In this sense, $\mathcal{N}4$ can be seen as a paraconsistent weakening of $\mathcal{N}3$. In fact, the logic $\mathcal{N}3$ is the explosive extension of $\mathcal{N}4$, i.e., the least logic in the appropriate language containing the axioms of $\mathcal{N}4$ and the law $\sim p \rightarrow (p \rightarrow q)$, from which we can derive $\mathcal{N}3 \vdash \perp \rightarrow p$ and $\mathcal{N}3 \vdash \sim \perp$ (where $\perp$ is defined as $\sim(p \rightarrow p)$) [Odi05].

7 Constructible Duality

We are interested here in the doctoral work of Patterson [Pat98] that provides a bridge between Nelson’s constructive systems and Dialectica/Chu spaces. Patterson recalls Nelson’s work on constructible falsity and then defines her own “constructible duality”. Constructible duality is obtained from constructible falsity by adding to the system exponential operators similar to the ones of Linear Logic, see page 8 of [Pat98]. The sequent rules for constructible duality CD are found in page 64, 65 of [Pat98].

Patterson discusses some algebraic models for constructible duality, including Chu spaces and dialectica categories predicate-free as above, but does not present an explicit Curry-Howard correspondence for the system. We can consider the category $\mathbf{LDial}_1(\mathbf{Hey})$, where objects are pairs of Heyting algebras together with a multivalued relation to $K = 1$, so triples (H, J, α) , where $\alpha : H \times J \rightarrow 1$. Maps are pairs of Heyting algebra morphisms (f_0, f_1) , $f_0 : H \rightarrow H_1$, $f_1 : J_1 \rightarrow J$, satisfying the dialectica condition. However, as suggested above, if $K = 1$ (where 1 is the terminal object), then the category $\mathbf{LDial}_1(\mathbf{Hey}) = \mathbf{Chu}_1(\mathbf{Hey})$, so these are simply pairs of Heyting algebras, with a positive-and-negative notion of morphism. The structures of $\mathbf{LDial}_K(\mathbf{Hey})$ and $\mathbf{Chu}_K(\mathbf{Hey})$ are different, as explained in [dP06], however over the terminal object, these structures become less important, as in the singleton category, all logical operations are the same.

The algebras considered by Patterson in [Pat98] do seem to correspond to instances of the Dialectica and Chu constructions when the ‘field’ K is the terminal object of the category in question. It would be sensible to try to provide for the logics of constructive falsity (either $\mathcal{N}3$, $\mathcal{N}4$, $\mathcal{S}$ or CD) a Curry-Howard correspondence.

This requires a Natural Deduction term calculus, with the appropriate properties and this necessitates a resolution of the main problem with Nelson’s systems, described by Tranchini¹ in [Tra21]

¹We provide an informal translation.

In short, in $\mathcal{N}4$ logic, the mutual implication of two formulas is not a congruence, and the principle of

as

En somme, dans la logique $\mathcal{N}4$, l'implication mutuelle de deux formules n'est pas une congruence, et le principe de substitution (commun à la logique intuitionniste, à la logique classique et à la plupart des systèmes logiques) :

$$\frac{A \iff B}{C[A/p] \iff C[B/p]}$$

est invalide, et doit être affaibli ainsi :

$$\frac{A \iff B \quad \sim A \iff \sim B}{C[A/p] \iff C[B/p]}$$

Thus providing a Curry-Howard interpretation for Nelson's logics is more complicated than usual. We cannot simply follow the usual procedure as described in 'Proofs and Types' [GTL89], for example.

Kamide and Wansing [KW12] suggest a solution to the problem, passing through some embedding of $\mathcal{N}4$ into substructural logics. Tranchini has a different proposal for a Curry-Howard term calculus [Tra18] which would be interesting to develop side by side with the corresponding categorical semantics. His proposal involves 'polarized formulas', but this requires some time and hence it is left for future work.

8 Exponentials for translations

When we discuss a categorical model of Intuitionistic Linear Logic (without modalities) we must provide a symmetric monoidal closed category with products and coproducts. This is sometimes called a **linear category**.

However, if we do have modalities in the model, we should end up with a (symmetric monoidal) adjunction, between the original linear category $\mathcal{L}$ and the cartesian closed category $\mathcal{C}$ (which models propositional intuitionistic logic). The adjunction models the modality "!", Girard's "of course!". The characteristics of the modality match very closely Girard's translation [Gir87]. And we know that while cartesian (closed or not) structures are unique up to isomorphism, monoidal closed structures are not unique. Similarly, the modalities themselves are not determined: there may be several different (not equiprobable) modalities on the same category. The traditional example is the category of relations. This has more than one usual modality: both the finite multisets or simply the full powerset can be used as a modality, as explained in https://www.pls-lab.org/en/Relational_model_of_Linear_Logic

One might want to consider Nelson's strong negation in this context of exponentials and translations between models, as it is done in [PC00].

In the category $\text{LDial}_1(\text{Hey})$ that we discussed before, we could consider an exponential such that

$$!A = A \& I = (U \times 1, X + 1, \alpha^*)$$

where $I = (1, 1, id : 1 \times 1 \rightarrow 1)$ and $\&$ denotes the categorical product in $\text{LDial}_1(\text{Hey})$. We have not discussed categorical products in $\text{LDial}_1(\text{Hey})$, but they are easy, the product of $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$ is simply the object $(U \times V, X + Y, \text{choose})$, where for the pair (u, v) to relate to an element of the disjoint sum $X + Y$, $(x, 0) + (y, 1)$, we simply choose the kind of element of the disjoint sum we have.

The idea of a simplified exponential $!$, which is just a product with the identity for tensor has been considered several times before. In [PC00] they have to ask for something else: one asks for the

substitution (common to intuitionist logic, classical logic, and most of the logical systems):

$$\frac{A \iff B}{C[A/p] \iff C[B/p]}$$

is invalid, and must be weakened as follows:

$$\frac{A \iff B \quad \sim A \iff \sim B}{C[A/p] \iff C[B/p]}$$

Linear Logic	Constructible Duality
0	0
$\top$	1
I	I
$\perp$	I
$A \otimes B$	$\neg((\mathcal{A} \supset \neg \mathcal{B}) \wedge (\mathcal{B} \supset \neg \mathcal{A}))$
$A \multimap B$	$(\mathcal{A} \supset \mathcal{B}) \wedge (\neg \mathcal{B} \supset \neg \mathcal{A})$
$A \wp B$	$(\neg \mathcal{A} \supset \mathcal{B}) \wedge (\neg \mathcal{B} \supset \mathcal{A})$
$A \& B$	$\mathcal{A} \wedge \mathcal{B}$
$A \oplus B$	$\mathcal{A} \vee \mathcal{B}$
$A^\perp$	$\neg \mathcal{A}$
!A	$\mathcal{A} \wedge I$
?A	$\mathcal{A} \vee I$

Table 1: Translating Linear Logic to Constructible Duality

existence of a map $A \multimap I$ and that $I^\perp = I$. Now asking for $I^\perp = I$ is easy, the linear negation here is simply swapping coordinates, so we do have $I^\perp = I$. But to provide a map $A \multimap I$ we must produce both a map $U \rightarrow 1$, (easy since 1 is a terminal object), but also a map $1 \rightarrow X$, which requires that X is well-pointed. If we ask for that, the calculations in [PC00] seem enough to provide us with a generalization of their theorem:

Theorem 8.1 (Patterson2000) *The categorical model of constructive duality $\text{LDial}_1(\text{Hey})$ is also a model of classical linear logic.*

Let A be a formula of Linear Logic. If A is valid in (classical) linear logic, then $\mathcal{A}$ is valid in constructible duality, where $\mathcal{A}$ is simply the translation described in the table above.

9 Conclusion

We discussed briefly how we can provide categorical semantics of some sort for Nelson system of constructible duality, based on Patterson’s work and on the dialectica construction. Patterson used Chu spaces mostly, but in the case discussed here (where relations are evaluated in 1) there is no difference between Chu spaces and dialectica ones.

To conclude this note, I wanted to mention why I think the Curry-Howard correspondence for (some of) the Nelson systems is important. In many practical situations, both in natural language and in more abstract settings, we have a situation where counterexamples are easier to understand than the traditional intuitionistic negation with its eventual implication into a falsity. If we want logics that agree with our intuitions, we want systems that can cope with the naturality of these basic negations, at the same time as having the canonical logical properties that we have come to expect as necessary from logical systems: syntax and semantics aligned to provide soundness and completeness theorems, syntactic cut elimination and/or normalization, some sort of subformula property, etc. It would be good if the extra benefits pointed out by explicit programming [Pat98] and its use of fixed points could be harnessed as well, but even without this extra computational layer, having a Nelson system that actually provided a constructive negation, even at the price of two implications (one weak, one strong), seems a goal worth working for.

References

- [AN84] Ahmad Almukdad and David Nelson. Constructible Falsity and Inexact Predicates. *The Journal of Symbolic Logic*, 49(1):231–233, 1984.
- [CGM⁺24] Matteo Capucci, Bruno Gavranović, Abdullah Malik, Francisco Rios, and Jonathan Weinberger. On a fibrational construction for optics, lenses, and dialectica categories, 2024.

- [dP89a] V C V de Paiva. The Dialectica Categories. In *Categories in Computer Science and Logic: Proceedings of the AMS-IMS-SIAM Joint Summer Research Conference Held June 14-20, 1987 with Support from the National Science Foundation*, volume 92, page 47. American Mathematical Soc., 1989.
- [dP89b] Valeria CV de Paiva. A dialectica-like model of linear logic. In *Category Theory and Computer Science*, pages 341–356. Springer, 1989.
- [dP91a] Valeria de Paiva. Categorical multirelations, linear logic and petri nets, computer lab technical report. 1991.
- [dP91b] Valeria Correa Vaz de Paiva. The dialectica categories. Technical report, University of Cambridge, Computer Laboratory, 1991.
- [dP02] Valeria de Paiva. Lineales: algebras and categories in the semantics of linear logic. In *Words, Proofs, and Diagrams*(eds. D. Barker-Plummer, D. Beaver, Johan van Benthem and P. Scotto di Luzio. CSLI, 2002.
- [dP06] Valeria de Paiva. Dialectica and chu constructions: cousins? *Theory and Applications of Categories*, 17:127–152, 2006.
- [GF⁺86] K. Gödel, S. Feferman, et al. *Kurt Gödel: Collected Works: Volume II: Publications 1938-1974*, volume 2. Oxford University Press, 1986.
- [Gir87] Jean-Yves Girard. Linear logic. *Theoretical computer science*, 50(1):1–101, 1987.
- [GTL89] Jean-Yves Girard, Paul Taylor, and Yves Lafont. *Proofs and Types*. Cambridge University Press, USA, 1989.
- [Hof11] Pieter Hofstra. The dialectica monad and its cousins. *Models, logics, and higherdimensional categories: A tribute to the work of Mihály Makkai*, 53:107–139, 2011.
- [Kle52] Stephen Cole Kleene. Introduction to metamathematics. 1952.
- [KW12] Norihiro Kamide and Heinrich Wansing. Proof theory of Nelson’s paraconsistent logic: A uniform perspective. *Theoretical Computer Science*, 415:1–38, 2012.
- [Mos22] Sean Moss. Dependent products of polynomials, talk at the polynomials workshop ii, 2022.
- [MvG18] Sean K. Moss and Tamara von Glehn. Dialectica models of type theory. In *Proceedings of the 33rd Annual ACM/IEEE Symposium on Logic in Computer Science*. ACM, jul 2018.
- [Nel49] David Nelson. Constructible Falsity. *The Journal of Symbolic Logic*, 14(1):16–26, 1949.
- [Nel59] David Nelson. Negation and separation of concepts in constructive systems. *Constructivity in mathematics*, 39:205–225, 1959.
- [Niu] Nelson Niu. Dialectica categories and polynomial functors (part 1).
- [NRMS20] Thiago Nascimento, Umberto Rivieccio, João Marcos, and Matthew Spinks. Nelson’s logic S. *Logic Journal of the IGPL*, 28(6):1182–1206, 05 2020.
- [Odi04] Sergei P. Odintsov. On the representation of n4-lattices. *Studia Logica*, 76:385–405, 2004.
- [Odi05] Sergei P. Odintsov. The class of extensions of Nelson’s Paraconsistent Logic. *Studia Logica: An International Journal for Symbolic Logic*, 80(2/3):291–320, 2005.
- [OdPPR18] Carlos Olarte, Valeria de Paiva, Elaine Pimentel, and Giselle Reis. The ILLTP library for intuitionistic linear logic. In Thomas Ehrhard, Maribel Fernández, Valeria de Paiva, and Lorenzo Tortora de Falco, editors, *Proceedings Joint International Workshop on Linearity & Trends in Linear Logic and Applications, Linearity-TLLA@FLoC 2018, Oxford, UK, 7-8 July 2018*, volume 292 of *EPTCS*, pages 118–132, 2018.

- [Pat98] Anna Patterson. The logic of constructible duality (phd thesis), 1998.
- [PC00] Anna Patterson and Tom Costello. Exponentials as projections from paraconsistent logics. In *Frontiers of Paraconsistent Logic, Research Studies Press Ltd., Baldock, Hartfordshire, England, pages 209–222*, 2000.
- [Pra99] Vaughan Pratt. Chu spaces. *School on Category Theory and Applications (Coimbra, 1999)*, 21:39–100, 1999.
- [Shu18] Michael Shulman. The 2-chu-dialectica construction and the polycategory of multivariable adjunctions. 2018.
- [SN] David I. Spivak and Nelson Niu. Polynomial functors: A general theory of interaction.
- [Tra18] Luca Tranchini. Curry-Howard for N4, personal communication, 2018.
- [Tra21] Luca Tranchini. Negation. In *Précis de philosophie de la logique, eds F. Poggiolesi and P. Wagner*. Editions de la Sorbonne, Paris, 2021.